Transfer: waves and signals

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose appropriate wave models and explain what a signal establishes.

SUCCESS CRITERIA - I can...

1. Use wave speed and round-trip geometry correctly.
2. Separate frequency, amplitude and direction effects.
3. State the limits of an inference from a signal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A radar echo's complete outward-and-return path is 5000 m. Find the target range and explain the division.

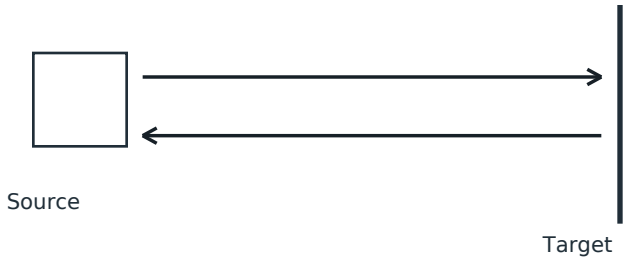
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v = f\lambda$; $T = 1/f$; $d_{range} = v\,\Delta t/2$.
d_{range} is one-way range from round-trip time. Match speed and model to the wave and medium.

Worked reasoning: An echo time measures range using travel speed, while a frequency shift can indicate radial motion. Different observations answer different questions about the same target.

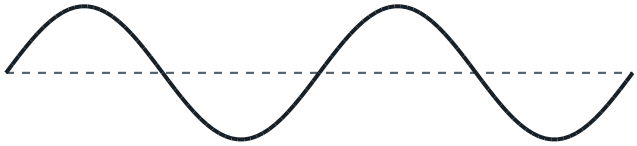
Your turn: Draw a measurement plan labeling outgoing pulse, return pulse and the time interval.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Sketch a wave and label amplitude and wavelength 0. Explain which tells about oscillation size and which about spatial repetition.

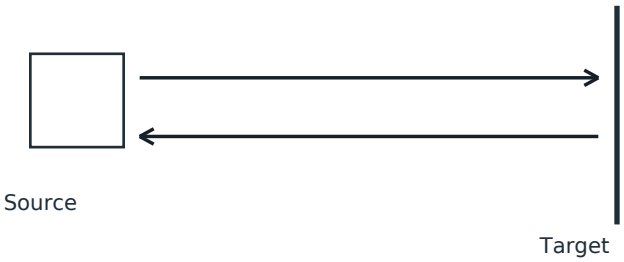


Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A wave simulator uses frequency 5 Hz and wavelength 2 m. Find propagation speed. An echo using that same speed returns after 4 s. Label the round trip and calculate the target range.



Marzano check - circle: ① ② ③

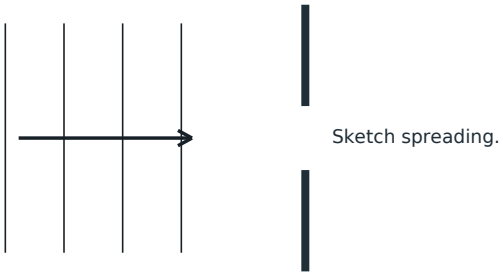
05 Explain your reasoning

A source gets louder while its frequency remains fixed. Does pitch necessarily rise? Explain using wave properties.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

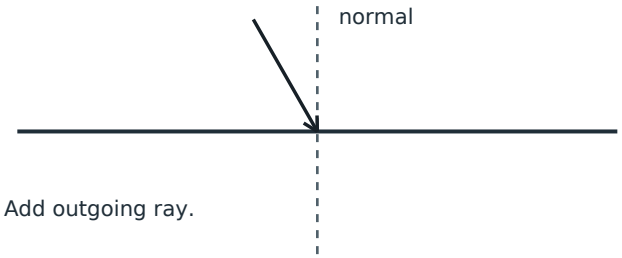
Two waves meet the same opening with wavelengths 4 cm and 6 cm. Which generally diffracts more? Explain the relevant comparison.
Label wavelength and opening width; sketch the emerging wavefronts.



Marzano check - circle: 1 2 3

07 Exit check

A ray enters a slower medium at an angle. Sketch whether it bends toward or away from the normal and name what happens to frequency.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A sound echo arrives after 0.5 s and a light echo after 2 microseconds. Use 340 m/s and 3×10^{8} m/s. Which target is farther?



Marzano check - circle: 1 2 3